

1. (Revision on first-year calculus)

- (a) Find the positive number(s) x for which the distance between the points $A(1, -1, x)$ and $B(-2x, 3, 1)$ is equal to $\sqrt{19}$.
- (b) Find the cartesian equation of the plane through the point $(1, -1, 3)$ and parallel to the plane with cartesian equation $3x + y + z = 7$.
- (c) Find an equation satisfied by the coordinates of the point $P(x, y, z)$ lying at equal distance from the points $A(-1, 3, 2)$ and $B(1, 0, 4)$.
- (d) Write down the equation of a sphere with centre (x_0, y_0, z_0) and radius r .
- (e) Find the equation of a sphere with one of its diameters having end-points $A(-3, 2, -5)$ and $B(1, 4, -1)$.
- (f) Find the centre and radius of the sphere whose equation is

$$x^2 + 6x + y^2 + 4y + z^2 - z + \frac{45}{4} = 0.$$

2. Find the domain of the vector function $\vec{r}(t) = \left(\frac{1}{\arccos(t-2)}, \arctan(t+1), \ln|2t-3| \right)$.

3. Find the limit $\lim_{t \rightarrow 0^+} \vec{r}(t)$ if

$$(a) \quad \vec{r}(t) = \left(\frac{1}{\pi} \arctan\left(\frac{1}{t}\right), -\sqrt{t} \sin\left(\frac{1}{t^2}\right), e^t \right); \quad (b) \quad \vec{r}(t) := \left(\frac{\sin(2t)}{t}, t \ln t, \sqrt{t} \sin^2\left(\frac{1}{t^2}\right) \right).$$

4. For each of the following, sketch the curve in $\mathbb{R}^2$ parameterized by:

$$(i) \quad \vec{r}(t) = (\cos t, \sin t), \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}; \quad (ii) \quad \vec{r}(t) = (\sqrt{1 - (\ln t)^2}, \ln t), \quad e^{-1} \leq t \leq e.$$

What do you observe?

5. Let $\mathcal{C}$ be the curve described by the vector function $\vec{r}(t) := \left(\frac{1-t^2}{1+t^2}, e^t, \frac{2t}{1+t^2} \right)$.

- (i) Say whether the curve $\mathcal{C}$ lies on the cylinder $x^2 + z^2 = 1$.
- (ii) Evaluate the indefinite integral $\int \vec{r}(t) dt$.

6. At what point(s) does the curve described by the vector function $\vec{r}(t) = (t^2 + 3, 2t^2, t - 1)$ intersect the plane with cartesian equation $2x - y - z = 3$?

7. Consider the curve $\mathcal{C}$ described by the vector function $\vec{r}: [0, \frac{3\pi}{4}] \rightarrow \mathbb{R}^2$ defined by

$$\vec{r}(t) = (3 \sin(2t), 2 \cos(2t)) \quad \text{for every } 0 \leq t \leq \frac{3\pi}{4}.$$

Sketch the curve $\mathcal{C}$ indicating the points of intersection with the axes.

8. Evaluate: (a) $\int_0^{\sqrt{\frac{\pi}{2}}} (t \sin(t^2), 2t^3, \sqrt{2}) dt$; (b) $\int \frac{1}{1+t^2} (1, t, t^2) dt$; (c) $\int_0^{\frac{\pi}{4}} (\sec t, \tan t, t \cos t) dt$.